# ENTERPRISE DATA GOVERNANCE FRAMEWORK DESIGN & IMPLEMENTATION FOR E-COMMERCE PLATFORM TRANSFORMATION

## DATA GOVERNANCE TARGET OPERATING MODEL (TOM)

### 1. EXECUTIVE SUMMARY

The Data Governance Target Operating Model (TOM) defines the structure, roles, processes, and technologies required to manage data as a strategic asset across the enterprise. It ensures alignment between business objectives, regulatory requirements, and data management practices.

### 2. OBJECTIVES

- Establish clear accountability for data ownership and stewardship
- Improve data quality and trust across all business domains
- Enable regulatory compliance and auditability
- Standardize data governance processes across the enterprise
- Support scalable governance aligned with modern data platforms

### 3. OPERATING MODEL OVERVIEW

#### 3.1. Governance Model Type

✓ **Federated Governance Model**

- Central governance defines policies & standards
- Domain teams execute governance within business units

### 4. GOVERNANCE STRUCTURE

#### 4.1. Organizational Layers

**Executive Layer**

- Chief Data Officer (CDO)
- Data Governance Council

**Strategic Layer**

- Data Governance Lead
- Domain Data Owners

**Operational Layer**

- Data Stewards
- Data Custodians (IT)

**Supporting Functions**

- Risk & Compliance
- Legal
- Security

### 4.2. Roles & Responsibilities

| Role | Responsibility |
|---|---|
| Data Owner | Accountable for data domain quality & compliance |
| Data Steward | Ensures data quality, metadata, definitions |
| Data Custodian | Technical management of data |
| Governance Council | Decision-making & policy approval |

## 5. RACI FRAMEWORK

| Activity | Owner | Steward | Custodian | Governance Council |
|---|---|---|---|---|
| Data Definition | A | R | C | I |
| Data Quality Monitoring | A | R | C | I |
| Access Control | A | C | R | I |
| Policy Approval | R | C | I | A |
| Metadata Management | A | R | C | I |
| Issue Resolution | A | R | C | I |

(R = Responsible, A = Accountable, C = Consulted, I = Informed)

## 6. CORE GOVERNANCE PROCESSES

### 6.1. Data Quality Management

- Define critical data elements (CDE)

  Determination criteria:
  - Regulatory/reporting impact
  - Revenue/payment impact
  - Customer experience impact
  - Fraud/risk impact
  - Operational dependency across domains

  Example of e-commerce CDE: customer_id, email, order_id, payment_status, sku, vendor_id.

- Establish data quality rules & thresholds
- Monitor via dashboards
- Issue management workflow:
  - Detect → Log → Assign → Resolve → Validate → Close

### 6.2. Metadata Management

- Maintain enterprise data catalog
- Define business glossary
- Track data lineage (source → transformation → consumption)

### 6.3. Data Access & Security

- Role-Based Access Control (RBAC)
- Attribute-Based Access Control (ABAC) (optional advanced)
- Access request & approval workflow
- Periodic access review

### 6.4. Data Lifecycle Management

- Define lifecycle stages:
  - Create → Store → Use → Archive → Delete
- Retention policies based on regulatory requirements

### 6.5. Data Classification

- Define classification levels:
    - Public
    - Internal
    - Confidential
    - Restricted (PII / highly sensitive)

## 7. GOVERNANCE ARTEFACTS

**Mandatory Artefacts:**

- Data Governance Policy
- Data Standards
- Data Quality Rules
- Data Classification Policy
- Data Access Policy
- Data Dictionary / Glossary
- Data Lineage Documentation
- Governance Control Register

## 8. TECHNOLOGY & TOOLING

### 8.1. Data Governance Tools

- Data Catalog: Collibra / Alation / Atlan / Microsoft Purview
- Data Quality: Informatica / Great Expectations

### 8.2. Data Platforms

- Data Warehouse (Snowflake, BigQuery)
- Data Lake / Lakehouse (Databricks)

### 8.3. Integration

- Metadata ingestion from ETL pipelines
- Automated lineage tracking

## 9. INTEGRATION WITH DATA ARCHITECTURE

- Governance embedded in:
  - Data ingestion pipelines
  - Transformation layers
  - BI / reporting layer

- Controls applied at:
  - Data creation
  - Data storage
  - Data access

## 10. COMPLIANCE & RISK MANAGEMENT

- Align with regulatory frameworks (e.g., GDPR principles)
- Implement audit trails
- Periodic compliance assessment
- Risk identification and mitigation

## 11. KPIs & SUCCESS METRICS

| **Metric** | **Description** |
|---|---|
| Data Quality Score | >95% of data meeting quality standards |
| Issue Resolution SLA | >95% of issues are resolved within the SLA according to severity |
| Data Availability | >99.5% availability for Tier-1 datasets during business hours / agreed SLA |
| Metadata Completeness | >98% of critical datasets documented in the catalog |
| Access Review Completion | 100% quarterly completion for privileged roles |

**KPI Definitions:**

| KPI | Formula (Example) | Target Phase 1–2 | Target Steady State |
|---|---|---|---|
| DQ Score | % CDE records that pass all quality rules | ≥90% | ≥95% |
| Metadata Completeness | % critical data assets with owner, steward, definition, classification, lineage | ≥80% | ≥98% |
| Owner Coverage | % critical data assets with Data Owner | ≥95% | 100% |
| Issue Resolution SLA | % of issues are resolved within the SLA according to severity | ≥80% | ≥95% |
| Access Review Completion | % quarterly access reviews completed on time | ≥90% | 100% |

## 12. IMPLEMENTATION ROADMAP

**Phase 1: Assessment**

- Current state analysis
- Gap identification

**Phase 2: Design**

- Define TOM
- Develop policies & standards

**Phase 3: Pilot**

- Implement in selected domain

**Phase 4: Rollout**

- Scale across enterprise

**Phase 5: Optimization**

- Continuous improvement

## 13. CHANGE MANAGEMENT & ADOPTION

- Stakeholder engagement plan
- Training & awareness programs
- Governance champions per domain
- Communication strategy

## 14. ESCALATION FRAMEWORK

- Data issues escalated:
  - Steward → Owner → Governance Council
- SLA for resolution defined

  SLA Table (Example):

| Severity | Initial Response | Target Resolution |
|---|---|---|
| Critical | < 4 jam | < 24 jam |
| High | < 1 hari kerja | < 5 hari kerja |
| Medium | < 2 hari kerja | < 10 hari kerja |
| Low | < 5 hari kerja | < 30 hari kerja |

## 15. REVIEW & CONTINUOUS IMPROVEMENT

- Quarterly governance review
- Annual TOM update
- Continuous KPI monitoring

## 16. APPENDICES

### Appendix A: Data Quality Dimensions

- Accuracy
- Completeness
- Consistency
- Timeliness

**Appendix B: Domain-Centric Governance**

| Domain | Data Owner | Steward | Custodian |
|---|---|---|---|
| Customer | Head of CRM/Marketing | CRM Steward | Data Platform/IT |
| Product | Merchandising Lead | Catalog Steward | IT |
| Order & Transaction | Operations/Payments Lead | Order Steward | IT |
| Vendor | Supply Chain Lead | Vendor Steward | IT |

**Appendix C: Governance Cadence**

| Cadence | Forum | Purpose |
|---|---|---|
| Monthly | Domain Governance Review | DQ trends, open issues, access exceptions |
| Quarterly | Governance Council | Policy decisions, cross-domain conflicts, KPI review |
| Annually | Policy/TOM Review | Regulatory changes, roadmap reprioritization |

**Appendix B: Sample Workflow**

- Data Issue Management Workflow Diagram

## Visual Diagram (Architecture + Governance Flow)